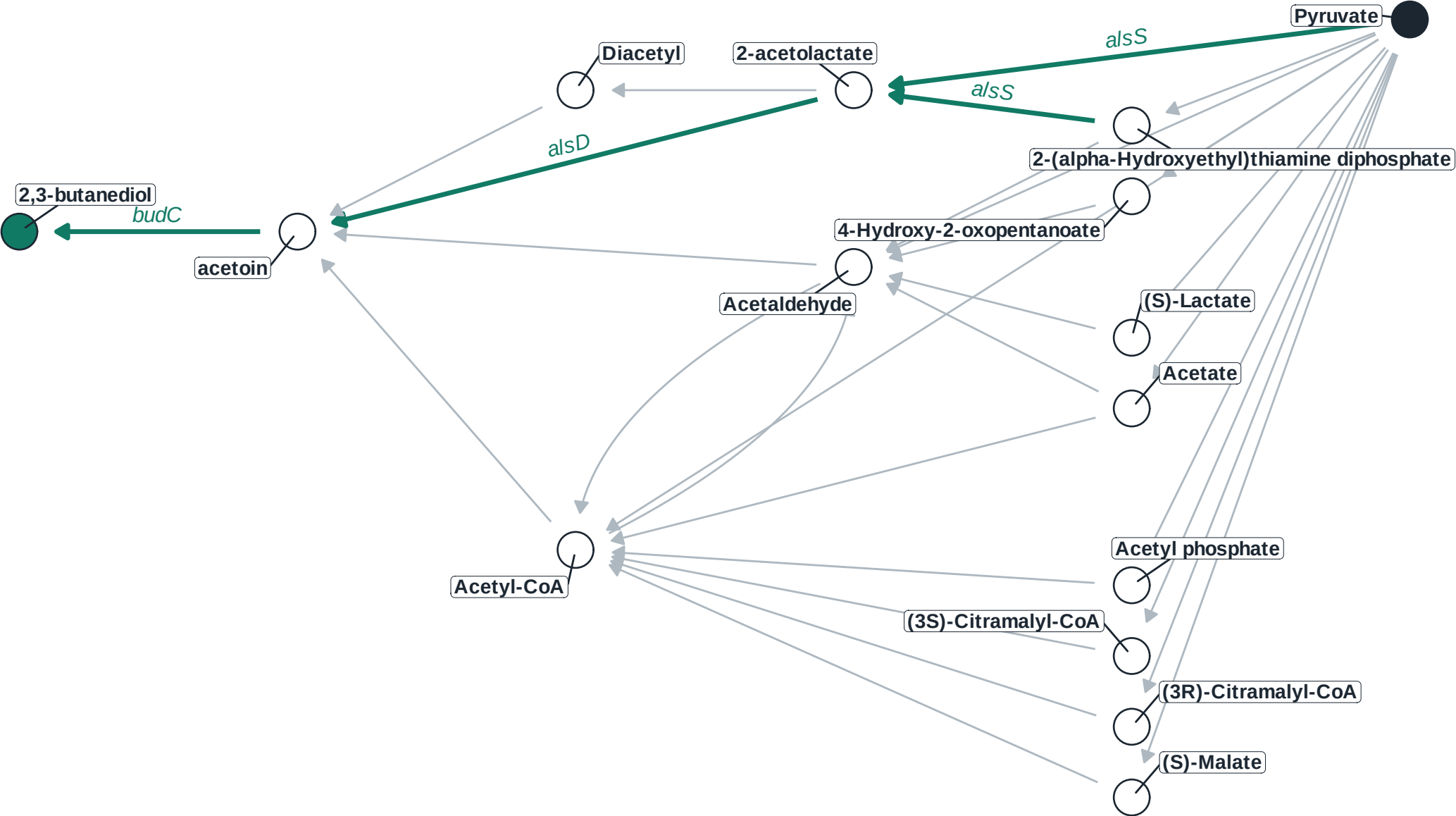


All native routes from pyruvate to 2,3-butanediol found by PanRoute

carbon-skeleton retrosynthetic search over KEGG · 16 routes · stereoisomers merged · **canonical route highlighted**

→ alternate route ➡ canonical: alsS → alsD → budC



Nodes = metabolites; directed edges = enzymatic steps (biological gene symbols). All routes converge on acetoin, the committed precursor of 2,3-butanediol (final step budC). Alternate routes are other carbon-skeleton paths the search found to reach that convergence.